

REVIEW

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Artificial intelligence applications and pedagogical challenges in music education

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Abstract

This mini review synthesizes recent advancements in the integration of artificial intelligence (AI) within instrumental music education, emphasizing both computational methods and pedagogical frameworks. Drawing from the top 50 highly cited Scopus-indexed documents, the review identifies dominant AI techniques such as deep learning, transformer architectures, and generative models. These technologies enhance practice efficiency, personalize instruction, and improve assessment objectivity. However, challenges persist, including dataset bias, limited cultural sensitivity, and constraints in expressive feedback. Thematic and technical analyses reveal a strong focus on composition and performance domains, with creativity and feedback as key pedagogical impacts. The review integrates pedagogical models such as TPACK, SAMR, and Bloom's taxonomy to contextualize AI adoption. Findings suggest that hybrid models combining AI analytics with human instruction offer the greatest educational value. Future research should prioritize culturally adaptive systems, ethical transparency, and inclusive design to ensure equitable and meaningful integration of AI in music pedagogy.

Keywords Artificial intelligence, Music education, Pedagogical efficacy, Technological integration, Assessment methodologies

1 Introduction

Artificial intelligence (AI) is rapidly reshaping the educational landscape by introducing data-driven, adaptive, and personalized learning environments across disciplines. In recent years, AI has evolved from a supplementary tool to a transformative force in pedagogy, enabling educators to address long-standing instructional challenges with unprecedented precision. Within this broader context, music education, particularly instrumental instruction has begun to benefit from AI's analytical and feedback capabilities. Traditional music pedagogy often struggles with providing timely, individualized feedback during practice sessions, a gap that AI technologies are increasingly able to fill. Emerging innovations such as real-time audio analysis, machine learning algorithms, and intelligent tutoring systems offer scalable solutions that enhance both the technical and expressive dimensions of music learning. These developments mark a significant



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shift in how musical competencies are cultivated, assessed, and supported in contemporary educational settings.

Recent advancements in computational models such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), Bidirectional LSTM (Bi-LSTM) with attention mechanisms, and Reinforcement Learning (RL) have begun to address persistent challenges in instrumental instruction, including the need for real-time feedback, accurate performance assessment, and adaptive learning environments [18, 46]. Modern AI platforms are now capable of analyzing key musical elements such as pitch accuracy, rhythmic precision, articulation, and tonal quality [70, 77, 92, 129, 136]. These systems not only identify performance errors but also provide learners with targeted and actionable feedback, enabling students to refine their skills independently, outside of formal instruction [1, 55]. For example, Bi-LSTM models have achieved up to 91.9% accuracy in music performance assessment, and hybrid CNN-LSTM models report classification accuracies of over 99% for recognizing violin bowing techniques [29, 46]. As a result, learners often report increased autonomy, stronger motivation, and improved self-efficacy, outcomes closely associated with long-term retention and musical growth [65, 66].

These technological advancements have prompted educators and researchers to explore how established pedagogical frameworks can be adapted to AI-enhanced environments. The TPACK framework, which integrates technological, pedagogical, and content knowledge, has shown strong relevance in music education. Bauer [13] developed an instrument to measure music educators' TPACK and found a positive correlation between TPACK proficiency and effective technology integration, suggesting that educators with higher TPACK are better positioned to adopt AI tools in instrumental instruction. Similarly, Bloom's taxonomy has been used to guide AI-driven assessment systems. Gavhane and Pagare [40] demonstrated how deep learning models can automate cognitive level classification, offering scalable solutions for evaluating musical creativity and performance. Similarly, through exploration, navigation, and interaction within such environments, students are further supported in developing their cognitive and spatial abilities [48]. In this sense, the findings extend existing understanding by illustrating how specific characteristics of AR meaningfully contribute to cognitive engagement and spatial reasoning, thereby exerting a substantial influence on the overall teaching and learning process [102].

Next, Patil et al. [90] further explored automated question classification aligned with Bloom's levels, indicating potential for adaptive feedback in music learning environments. The SAMR model provides a lens to evaluate the transformative impact of AI tools in education. Rabu et al. [95] applied SAMR and Bloom's taxonomy to assess ChatGPT's integration in classroom instruction, highlighting its capacity to support higher-order thinking and redefine learning tasks. Although direct applications in music education remain limited, these frameworks offer valuable insights for designing AI-supported pedagogical strategies that balance innovation with cognitive rigor. Following recent pedagogical advancements, Tariq et al. [117] emphasized the role of project-based learning in enhancing computational thinking and sustainable development in STEM education, offering a transferable framework for music pedagogy.

A conceptual model proposed by Lee and Lee [64], shown in Fig. 1 below, illustrates how AI can support the development of musical competencies through structured,

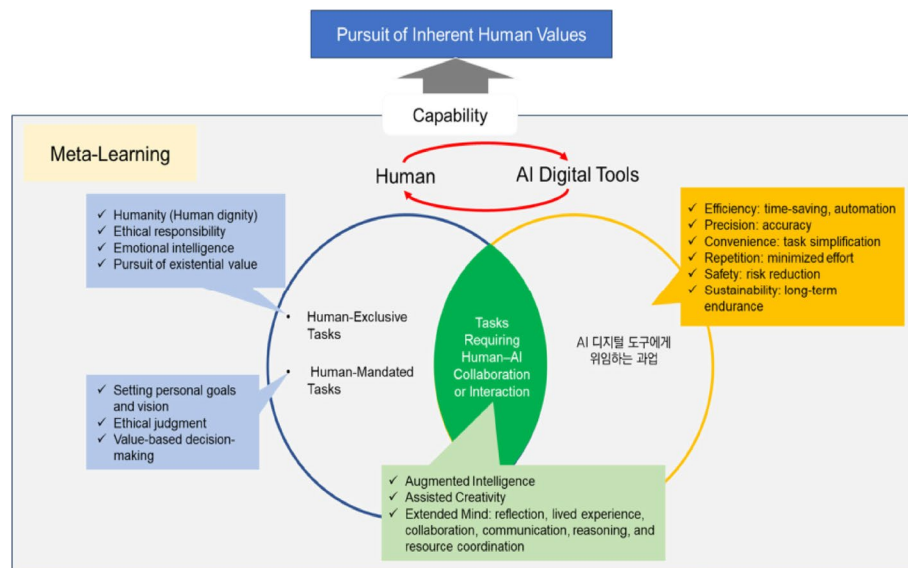


Fig. 1 Conceptual model of Intelligent Problem-solving Learning.

Source: Lee and Lee [64]

feedback-driven learning processes. The model emphasizes the interaction between human values, AI digital tools, and collaborative learning tasks. It highlights how AI can enhance efficiency, creativity, and accessibility while supporting learners in setting personal goals and applying effective learning strategies [68]. At the center of this model is the concept of meta-learning, which reflects the integration of human dignity and technological intelligence in pursuit of meaningful educational outcomes.

Despite these advances, significant technical and practical challenges remain. Current AI systems often struggle with processing complex, multimodal data (audio, visual, symbolic), extracting salient features, and generalizing algorithms across diverse educational scenarios [92, 135]. Ensuring low latency for real-time feedback is essential for music learning, and although gesture recognition and experiential teaching systems have made progress, further optimization is required [54, 55]. Additional concerns include ethical and cultural biases in datasets, limitations in feature extraction, and the need for explainable, equitable, and robust AI architectures [22, 23].

While the literature highlights several pedagogical benefits of AI, including improved student engagement, more efficient practice routines, and enhanced learning outcomes, there are still important gaps that need to be addressed. These include concerns about the scalability of AI tools, ethical considerations related to data use and bias, and the evolving role of educators in AI-supported learning environments [82, 97, 103, 106]. This mini review synthesizes the latest advances in AI algorithms and system architectures for music education, critically examining their pedagogical impact, technical limitations, and future research directions. By bridging the perspectives of computer science and music pedagogy, the review aims to guide researchers, developers, and educators in leveraging AI to enrich music learning while addressing the complex technical and ethical challenges that define this interdisciplinary field.

2 Methods

This mini review was conducted using the Scopus database exclusively, based on methodological justifications supported by recent literature. Scopus is one of the largest curated abstract and citation databases, offering extensive global and regional coverage of scientific journals, conference proceedings, and books. Its rigorous content selection process, overseen by an independent Content Selection and Advisory Board, ensures the inclusion of only high-quality publications, thereby enhancing the reliability of systematic and mini reviews [9].

Scopus also provides enriched metadata and comprehensive author and institutional profiles, supported by advanced profiling algorithms and manual curation. These features contribute to high precision and recall in bibliometric and literature analyses [9]. Comparative studies have shown that Scopus offers broader content coverage and more user-friendly features than other databases such as Web of Science and Google Scholar, making it particularly suitable for interdisciplinary research [10, 81]. Additionally, Scopus has been shown to improve search efficiency significantly, with one study reporting a 62.5% time saving compared to traditional manual methods [19].

Although Web of Science (WoS) and Google Scholar were considered, Scopus was selected due to its superior coverage, metadata quality, and compatibility with bibliometric tools. Compared to WoS, Scopus includes a broader range of conference proceedings and non-English literature, particularly from Eastern and Western Europe [8, 47, 61]. While Google Scholar offers extensive coverage, its citation metrics are often inconsistent and less frequently updated, limiting its reliability for bibliometric analysis [14, 31, 36]. Scopus also provides approximately 20% more citation coverage than WoS in certain fields [36], and its integration with tools like BiblioMagika v2.10.3 further enhances research efficiency [10, 11]. From the initial pool of 255 records retrieved through Scopus, 165 were retained after applying inclusion and exclusion criteria. To ensure analytical depth and citation relevance, a final subset of 50 highly cited and thematically aligned documents was selected for detailed bibliometric and thematic analysis.

2.1 Search strategy

The literature search was conducted on 29 September 2025, utilizing a structured Boolean query applied to the Title, Abstract, and Keywords fields within the Scopus database. This approach was chosen to ensure a focused and comprehensive retrieval of relevant studies at the intersection of artificial intelligence and music education, with particular emphasis on pedagogical outcomes and instructional design.

The Boolean query was carefully constructed to include a wide range of AI-related terms such as “artificial intelligence,” “machine learning,” “deep learning,” “generative AI,” and “intelligent tutoring systems,” combined with music education-specific keywords like “instrumental music,” “music pedagogy,” “music learning,” and “music composition.” To capture the educational impact, additional terms such as “assessment,” “feedback,” “creativity,” “student engagement,” and “digital didactics” were included.

To maintain the quality and relevance of the results, the search was limited to English-language publications and filtered by document type (articles and conference papers) and source type (journals and proceedings). The final query string was:

TITLE-ABS-KEY (("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR "intelligent tutoring systems") AND ("music education" OR "instrumental music" OR "music pedagogy" OR "music learning" OR "music composition") AND ("assessment" OR "feedback" OR "creativity" OR "student engagement" OR "digital didactics")) AND (*LIMIT-TO* (*LANGUAGE*, "English")) AND (*LIMIT-TO* (*DOCTYPE*, "cp") OR *LIMIT-TO* (*DOCTYPE*, "ar")) AND (*LIMIT-TO* (*SRCTYPE*, "j") OR *LIMIT-TO* (*SRCTYPE*, "p")).

The review process adhered to the PRISMA 2020 guidelines, which are widely recognized for promoting transparency, reproducibility, and methodological rigor in systematic reviews. By following this structured approach, the study ensured that the selection of literature was both comprehensive and thematically relevant to the intersection of artificial intelligence and music education. The flow diagram presented in Fig. 2, adapted from Zakaria et al. [143] and Moher et al. [80], visually represents the stages of identification, screening, eligibility, and inclusion of sources. The search strategy was executed within the Scopus database, chosen for its extensive coverage and metadata quality. The Boolean query was applied to the Title, Abstract, and Keywords fields, targeting publications from 2006 to 2025, written in English, and limited to journal articles and conference papers. This ensured a focused retrieval of high-impact studies across diverse subject areas, including computer science, education, engineering, and the arts.

The diagram also reflects the scope and refinement process, including keyword optimization, thematic relevance filtering, and citation impact assessment. This multi-layered screening ensured that only the most pertinent and methodologically sound studies were included in the final synthesis. The inclusion of this diagram not only supports methodological transparency but also reinforces the interdisciplinary nature of the review, bridging technical AI developments with pedagogical applications in music education. Scopus was selected as the sole data source because of its extensive coverage, reliable metadata, and compatibility with BiblioMagika for bibliometric analysis. Prior studies have shown that Scopus provides broader coverage than Web of Science, including more conference proceedings and journals from diverse publishers [9, 36, 81, 146]. Its proven comprehensiveness in bibliometric studies and inclusion of high-impact journals across disciplines provide confidence in the representativeness of the dataset used in this review.

2.2 Screening and selection

The initial search yielded a total of 255 records, which were subjected to a multi-stage screening process to ensure thematic relevance and methodological rigor. Studies were considered eligible if they explicitly addressed the application of artificial intelligence in music education, with a clear pedagogical or technical focus. Exclusion criteria included off-topic content, lack of methodological transparency, duplicate entries, and incomplete metadata. Although the use of a single database (Scopus) minimizes duplicate records, the exported results were manually checked to ensure no duplicates were present. To enhance consistency and reliability, a data cleaning and harmonization procedure was implemented, involving the standardization of author names, institutional affiliations, and keyword descriptors. This refinement process also included citation impact assessment and thematic clustering to prioritize high quality and influential studies [98]. From the initial pool of 255 records retrieved through Scopus, 165 were retained after applying

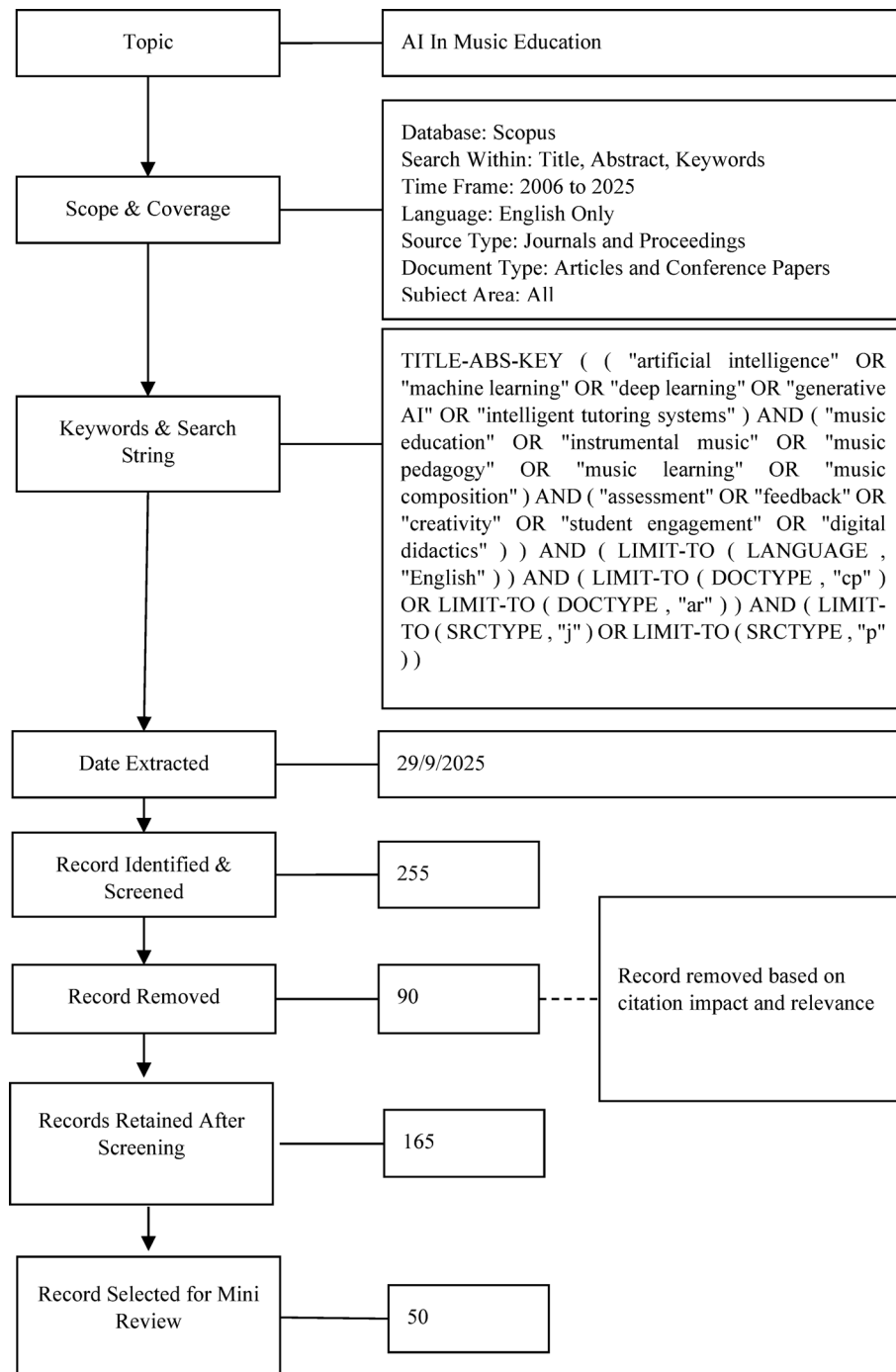


Fig. 2 Flow diagram of the search strategy.

Source: Zakaria et al. [143], Moher et al. [80]

inclusion and exclusion criteria. To ensure the analytical depth and citation-impact relevance required for a high-quality review, a final subset of the 50 most highly cited and thematically aligned documents was selected for detailed bibliometric and thematic analysis. This focused approach allows for a more rigorous examination of the most influential research at the intersection of AI and music education. The overall screening

and refinement process is summarized visually in Fig. 1, which outlines the number of records identified, removed, and retained at each stage.

2.3 Data analysis

To ensure a comprehensive and methodologically sound synthesis, a dual-layer analytical approach was adopted, utilizing both bibliometric and thematic methods on complementary datasets.

1. Bibliometric Analysis (Full Dataset: n = 165).

Bibliometric mapping was conducted on the full set of 165 eligible publications to identify macro-level field trends. This analysis was performed using BiblioMagika v2.10.3 [3], a specialized tool for keyword co-occurrence analysis, source distribution, and thematic clustering. The objectives were to:

- Identify dominant research trends and influential authors.
- Map the distribution of publications across subject areas and at a time.
- Visualize the conceptual structure of the field through keyword networks. Rather than relying solely on graphical visualizations, the findings were synthesized into structured tables (e.g., Tables 1, 2 and 3) and figures to provide a clear, reproducible overview of the research landscape.

2. Thematic Analysis (Focused Subset: n = 50).

For an in-depth, qualitative synthesis, the 50 most highly cited and thematically aligned documents underwent detailed manual thematic coding. This focused subset was selected to ensure the analytical depth and citation-impact relevance necessary for a high-quality review of the most influential work. The coding process was guided by established bibliometric frameworks and refined inductively through keyword co-occurrence patterns, focusing on five key analytical dimensions:

- AI technique used
- Music education domain
- Pedagogical impact

Table 1 Citation Metrics
Source: Generated by the author(s) using biblioMagika® [3]

Main Information	Data
Publication Years	2006–2025
Total Publications	165
Citable Year	20
Number of Contributing Authors	463
Number of Cited Papers	109
Total Citations	877
Citation per Paper	5.32
Citation per Cited Paper	8.05
Citation per Year	46.16
Citation per Author	1.89
Author per Paper	2.81
Citation sum within h-Core	757
h-index	15
g-index	24
m-index	0.750

Table 2 Subject Area

Source: Generated by the author(s) using biblioMagika® [3]

Subject Area	Total Publication	Percentage (%)
Computer Science	118	71.52%
Engineering	53	32.12%
Social Sciences	36	21.82%
Mathematics	31	18.79%
Decision Sciences	19	11.52%
Arts and Humanities	16	9.70%
Psychology	13	7.88%
Physics and Astronomy	8	4.85%
Materials Science	7	4.24%
Neuroscience	7	4.24%
Medicine	6	3.64%
Business, Management and Accounting	4	2.42%
Biochemistry, Genetics and Molecular Biology	2	1.21%
Chemical Engineering	2	1.21%
Energy	2	1.21%
Environmental Science	2	1.21%
Health Professions	2	1.21%
Economics, Econometrics and Finance	1	0.61%
Multidisciplinary	1	0.61%

- Reported limitations or gaps
- Ethical or cultural considerations

These dimensions were extracted primarily from abstracts and, where necessary, supplemented by a full-text review to ensure contextual accuracy. The resulting data were organized into a thematic matrix (Table 4) and a technical summary matrix (Table 5). Table 4 provides a conceptual overview of AI applications, while Table 5 complements it by detailing technical metrics such as model accuracy, latency, and implementation constraints where available. Together, this dual-layered analytical strategy, combining a broad bibliometric overview with a focused thematic deep-dive, offers a holistic perspective that bridges pedagogical insights with computational realities, enhancing the interpretive depth and rigor of the review.

3 Results

This section presents the bibliometric and thematic findings. The bibliometric profile (Tables 1, 2 and 3, Figs. 2, 3, 4 and 5) is based on the full dataset of 165 publications, providing an overview of the field. The subsequent in-depth thematic and technical analysis (Tables 4 and 5) is derived from the final subset of 50 highly cited documents, selected for their significant influence as detailed in the Methods section. All analyses were conducted using BiblioMagika v2.10.3.

3.1 Document profile

Table 1 summarizes the citation metrics of the selected documents. The dataset spans from 2006 to 2025, comprising 165 publications authored by 463 contributors. The total citation count is 877, with an average of 5.32 citations per paper and 8.05 citations per cited paper. The h-index of 15 and g-index of 24 indicate moderate citation impact across the field. The m-index of 0.75 suggests consistent scholarly output over time.

Table 3 Highly Cited Documents (Top 50 Selected from 165 Eligible Records)

Source: Generated by the author(s) using biblioMagika® [3]

No.	Author(s)	Title	Source Title	TC	C/Y
1	Suh et al. [113]	AI as Social Glue: Uncovering the Roles of Deep Generative AI during Social Music Composition	Conference on Human Factors in Computing Systems—Proceedings	81	16.20
2	Jin et al. [56]	A Style-Specific Music Composition Neural Network	Neural Processing Letters	43	7.17
3	Tigre Moura and Maw [118]	Artificial intelligence became Beethoven: how do listeners and music professionals perceive artificially composed music?	Journal of Consumer Marketing	40	8.00
4	Pati et al. [89]	Assessment of student music performances using Deep Neural Networks	Applied Sciences (Switzerland)	39	4.88
5	Cui [26]	Artificial intelligence and creativity: piano teaching with augmented reality applications	Interactive Learning Environments	36	12.00
6	Tubadji et al. [120]	Cultural proximity bias in AI-acceptability: The importance of being human	Technological Forecasting and Social Change	35	7.00
7	Giraldo et al. [43]	Automatic assessment of tone quality in violin music performance	Frontiers in Psychology	30	4.29
8	Uzumcu and Acil-mis [124]	Do Innovative Teachers use AI-powered Tools More Interactively? A Study in the Context of Diffusion of Innovation Theory	Technology, Knowledge and Learning	29	14.50
9	Liu and Ting [69]	Polyphonic accompaniment using genetic algorithm with music theory	2012 IEEE Congress on Evolutionary Computation	27	1.93
10	Sabath�� et al. [100]	Deep recurrent music writer: Memory-enhanced variational autoencoder-based musical score composition and an objective measure	Proceedings of the International Joint Conference on Neural Networks	25	2.78
11	Ferreira et al. [38]	Generating Music with Data: Application of Deep Learning Models for Symbolic Music Composition	Applied Sciences (Switzerland)	22	7.33
12	Carey et al. [16]	Enhancing learning and teaching in the tertiary music studio through reflection and collaboration	Music Education Research	19	2.38
13	Ting et al. [119]	A novel automatic composition system using evolutionary algorithm and phrase imitation	IEEE Systems Journal	19	2.11
14	Zhou and Kim et al. [145]	Innovative music education: An empirical assessment of ChatGPT-4's impact on student learning experiences	Education and Information Technologies	18	9.00
15	Eremenko et al. [35]	Performance assessment technologies for the support of musical instrument learning	Canadian Society for Mechanical Engineering (CSME) International Congress	18	3.00
16	Wang et al. [128]	Soloist: Generating mixed-initiative tutorials from existing guitar instructional videos through audio processing	Conference on Human Factors in Computing Systems—Proceedings	15	3.00
17	Baudouin [12]	A reconstruction of Stria	Computer Music Journal	15	0.79
18	Chu et al. [24]	An Empirical Study on How People Perceive AI-generated Music	International Conference on Information and Knowledge Management, Proceedings	13	3.25
19	Bougueng et al. [15]	Calliope: A Co-creative Interface for Multi-Track Music Generation	ACM International Conference Proceeding Series	13	3.25
20	No-velli and Proksch [88]	Am I (Deep) Blue? Music-Making AI and Emotional Awareness	Frontiers in Neurobotics	12	3.00
21	Merch��n S��nchez-Jara et al. [79]	Artificial Intelligence-Assisted Music Education: A Critical Synthesis of Challenges and Opportunities	Education Sciences	11	5.50

Table 3 (continued)

No.	Author(s)	Title	Source Title	TC	C/Y
22	Wang and Jiang [130]	Managing Students' Creativity in Music Education—the Mediating Role of Frustration Tolerance and Moderating Role of Emotion Regulation	Frontiers in Psychology	11	2.75
23	Lee and Liu [63]	Use of decision trees to evaluate the impact of a holistic music educational approach on children with special needs	Sustainability (Switzerland)	11	2.20
24	Tzimeas and Mangina [122]	Jazz Sebastian bach: A GA system for music style modification	2006 International Conference on Systems and Networks Communications (ICSNC'06)	11	0.55
25	Kumar et al. [62]	Automatic Music Generation System based on RNN Architecture	2022 2nd International Conference on Technological Advancements in Computational Sciences (ICTACS)	10	2.50
26	Tanberk and Tükel [116]	Style-Specific Turkish Pop Music Composition with CNN and LSTM Network	2021 IEEE 19th World Symposium on Applied Machine Intelligence and Informatics (SAMII)	10	2.00
27	Dadman et al. [28]	Toward Interactive Music Generation: A Position Paper	IEEE Access	9	2.25
28	Dalmazzo et al. [29]	Applying Deep Learning Techniques to Estimate Patterns of Musical Gesture	Frontiers in Psychology	9	1.80
29	Chang and Chen [17]	AntsOMG: A framework aiming to automate creativity and intelligent behavior with a showcase on cantus firmus composition and style development	Electronics (Switzerland)	9	1.50
30	Ford et al. [39]	Reflection Across AI-based Music Composition	C&C'24: Proceedings of the 16th Conference on Creativity and Cognition	8	4.00
31	Giraldo et al. [42]	Automatic assessment of violin performance using dynamic time warping classification	2018 26th Signal Processing and Communications Applications Conference (SIU)	8	1.00
32	Lopes et al. [72]	Combining rules and proportions: A multiobjective approach to algorithmic composition	2017 IEEE Congress on Evolutionary Computation (CEC)	8	0.89
33	Zhang et al. [144]	Transforming Music Education Through Artificial Intelligence: A Systematic Literature Review on Enhancing Music Teaching and Learning	International Journal of Interactive Mobile Technologies	7	3.50
34	Kim et al. [60]	An Overview of Automatic Piano Performance Assessment within the Music Education Context	International Conference on Computer Supported Education, CSEDU—Proceedings	7	1.75
35	Roig et al. [99]	A non-homogeneous beat-based harmony Markov model	Knowledge-Based Systems	7	0.88
36	Chen [20]	Unlocking the Beat: How AI Tools Drive Music Students' Motivation, Engagement, Creativity and Learning Success	European Journal of Education	6	6.00
37	Holster [52]	Augmenting Music Education through AI: Practical Applications of ChatGPT	Music Educators Journal	6	3.00
38	Lian [67]	An artificial intelligence-based classifier for musical emotion expression in media education	PeerJ Computer Science	6	2.00
39	Gonzalez and McMullen [45]	The Design of an Algorithmic Modal Music Platform for Eliciting and Detecting Emotion	2020 8th International Winter Conference on Brain-Computer Interface (BCI)	6	1.00
40	Samuel and Pilát, [104]	Composing Multi-Instrumental Music with Recurrent Neural Networks	Proceedings of the International Joint Conference on Neural Networks	6	0.86

Table 3 (continued)

No.	Author(s)	Title	Source Title	TC	C/Y
41	Cheng et al. [21]	Nonnegative matrix factorization for real time musical analysis and sight-reading evaluation	Proceedings—ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing	6	0.33
42	Xu [137]	Design and Development of Music Intelligent Education System Based on Artificial Intelligence	2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE)	5	2.50
43	Shi [108]	Research on the Innovation Path of Music Education in Higher Vocational Colleges and Universities in the Context of the New Era	Applied Mathematics and Nonlinear Sciences	5	2.50
44	Yang and Liu [140]	Evaluation of Comprehensive Services of an Online Learning Platform Based on Artificial Intelligence	International Journal of Emerging Technologies in Learning	5	1.25
45	Vass [125]	Musical Co-creativity and Learning—the Fluid Body Language of Receptive-Responsive Dialogue	Human Arenas	5	0.63
46	Venugopal and Madhusudan [127]	Feasibility of music composition using artificial neural networks	2017 International Conference on Computing Methodologies and Communication (ICCMC)	5	0.56
47	Dan [30]	Social robot assisted music course based on speech sensing and deep learning algorithms	Entertainment Computing	4	4.00
48	Wang and Li [132]	Assessment of the Relationship Between Music Students' Self-Efficacy, Academic Performance and Their Artificial Intelligence Readiness	European Journal of Education	4	2.00
49	Subbulakshmi et al. [111]	AI based Emotion Aware Music Composition using Spotify App	2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)	4	2.00
50	Fernando et al. [37]	Assessment of Human Emotional Responses to AI-Composed Music: A Systematic Literature Review	2024 International Research Conference on Smart Computing and Systems Engineering (SCSE)	4	2.00

3.2 Subject area distribution

As shown in Table 2, the majority of publications fall under Computer Science (71.52%), followed by Engineering (32.12%), Social Sciences (21.82%), and Mathematics (18.79%). Notably, Arts and Humanities account for only 9.70%, indicating a technology-centric focus in AI-related music education research.

3.3 Publication trends

Figure 3 below illustrates the annual distribution of publications and citations. A significant increase is observed in 2024 and 2025, reflecting the growing interest in generative AI tools and their application in educational contexts. Figure 4 further highlights the cumulative growth of publications over time, suggesting a steady expansion of the research landscape.

3.4 Highly cited documents

Table 3 presents the top 50 highly cited documents selected based on citation frequency and thematic relevance. These publications reflect a diverse range of AI applications in music education, including generative composition systems, performance assessment technologies, and pedagogical frameworks. The most cited article is “AI as Social Glue:

Table 4 Thematic matrix of 50 top selected documents

Source: Generated by the authors

Author and Year	AI Technique Used	Music Education Domain	Pedagogical Impact	Limitations or Gaps
Suh et al. [113]	Deep Learning	Composition	Motivation/Engagement	Not Specified
Jin et al. [56]	Long Short-Term Memory	Composition	Creativity	Not Specified
Tigre Moura and Maw [118]	Other/Not Specified	Composition	Creativity	Ethical Concerns
Pati et al. [89]	Deep Neural Networks	Performance	Feedback/Assessment	Not Specified
Cui [26]	Augmented Reality	Performance	Feedback/Assessment	Technical Limitations
Tubadji et al. [120]	Other/Not Specified	Composition	Creativity	Bias
Giraldo et al. [43]	Machine Learning	Performance	Feedback/Assessment	Technical Limitations
Uzumcu and Acilmis [124]	Other/Not Specified	Pedagogy	Motivation/Engagement	Not Specified
Liu and Ting [69]	Genetic Algorithm	Composition	Creativity	Not Specified
Sabath� et al. [100]	Variational Autoencoder	Composition	Creativity	Not Specified
Ferreira et al. [38]	Transformer Architecture	Composition	Creativity	Not Specified
Carey et al. [16]	Other/Not Specified	Pedagogy	Motivation/Engagement	Not Specified
Ting et al. [119]	Genetic Algorithm	Composition	Creativity	Not Specified
Zhou and Kim [145]	ChatGPT (Generative Pre-trained Transformer)	Pedagogy	Feedback/Assessment	Not Specified
Eremenko et al. [35]	Other/Not Specified	Performance	Feedback/Assessment	Technical Limitations
Wang et al. [128]	Other/Not Specified	Composition	Feedback/Assessment	Not Specified
Baudouin [12]	Other/Not Specified	Composition	Creativity	Not Specified
Chu et al. [24]	Deep Learning	Composition	Feedback/Assessment	Not Specified
Bougueng et al. [15]	Deep Learning	Composition	Creativity	Not Specified
Novelli and Proksch [88]	Other/Not Specified	Composition	Creativity	Ethical Concerns
Merch�n S�nchez-Jara et al. [79]	Other/Not Specified	Pedagogy	Creativity	Ethical Concerns
Wang and Jiang [130]	Other/Not Specified	Pedagogy	Autonomy/Self-efficacy	Not Specified
Zhang et al. [144]	Other/Not Specified	Pedagogy	Feedback/Assessment	Technical Limitations
Ford et al. [39]	Variational Autoencoder	Composition	Creativity	Not Specified
Holster [52]	ChatGPT (Generative Pre-trained Transformer)	Pedagogy	Feedback/Assessment	Ethical Concerns
Subbulakshmi et al. [111]	Machine Learning	Composition	Creativity	Technical Limitations
Xu [137]	Machine Learning	Pedagogy	Motivation/Engagement	Not Specified
Fernando et al. [37]	Deep Learning	Composition	Creativity	Bias
Shi [108]	Other/Not Specified	Pedagogy	Creativity	Access/Equity
Lian [67]	Radial Basis Function	Performance	Feedback/Assessment	Not Specified
Samuel and Pil�t [104]	Long Short-Term Memory	Composition	Creativity	Not Specified
Gonzalez and Mc-Mullen [45]	Machine Learning	Composition	Creativity	Ethical Concerns
Roig et al. [99]	Markov Model	Composition	Creativity	Not Specified

Table 4 (continued)

Author and Year	AI Technique Used	Music Education Domain	Pedagogical Impact	Limitations or Gaps
Cheng et al. [21]	Matrix Factorization	Performance	Feedback/Assessment	Technical Limitations
Tzimeas and Mangina [122]	Genetic Algorithm	Composition	Creativity	Not Specified

Uncovering the Roles of Deep Generative AI during Social Music Composition” by Suh and Youngblom [113], which received 81 citations and averaged 16.2 citations per year. Other influential works include Jin et al. [56] on style-specific music composition using neural networks, and Moura and Maw [118] on listener perceptions of AI-composed music. These documents collectively demonstrate the increasing scholarly attention toward AI-enhanced music education, particularly in areas such as composition, gesture recognition, student engagement, and ethical implications.

3.5 Keywords co-occurrence

Figures 5 and 6 present the top 20 author and index keywords. “Artificial Intelligence,” “Music,” “Music Education,” and “Music Composition” are the most frequently occurring terms. Technical keywords such as “Deep Learning,” “Machine Learning,” “Neural Networks,” and “Long Short-Term Memory” also appear prominently, indicating the prevalence of advanced computational models in the reviewed literature.

3.6 Regional distribution and representation

Based on author affiliations in the full bibliometric dataset (n = 165), most studies originated from Europe (≈ 28%), East Asia (≈ 23%), and North America (≈ 19%). In contrast, Southeast Asia (≈ 9%), the Middle East and South Asia (≈ 11%), and Latin America (≈ 5%) contributed relatively few publications, while Africa (<3%) remains largely under-represented. This uneven geographic distribution highlights the concentration of research activity in developed regions and the scarcity of studies from emerging educational contexts, reflecting broader publication disparities in AI-related education research.

3.7 Thematic matrix

Table 4 presents the thematic matrix of the top 50 highly cited documents. Each study is categorized by author and year, AI technique used, music education domain, pedagogical impact, and reported limitations or gaps. The most frequently applied AI techniques include deep learning, genetic algorithms, transformer architectures, and ChatGPT. Composition and performance are the dominant music education domains, while creativity and feedback or assessment are the most commonly reported pedagogical impacts. Limitations reported include technical constraints, ethical concerns, bias, and access or equity issues.

In addition to the thematic matrix presented in Table 4 above, a merged summary table (Table 5) was developed to enhance the interpretive depth of the Results section. This table synthesizes author contributions, AI model types, pedagogical impacts, limitations, and technical metrics such as accuracy and latency where available. While Table 4 offers a conceptual overview of AI applications in music education, Table 5 complements it by highlighting the technical characteristics and constraints of these models.

Table 5 Technical Summary of AI Models in Music Education (Top 50 Documents)

Source: Generated by the authors

No.	Authors	AI Model	Accuracy	Latency	Pedagogical Impact	Limitations
1	Chen	Not specified	Not reported	Not reported	Engagement, Feedback	Emotion
2	Dan	Not specified	Not reported	Not reported	Engagement, Feedback	Emotion
3	Wang and Li	Not specified	Not reported	Not reported	Creativity, Assessment	Implementation
4	Merchán Sánchez-Jara et al.	Not specified	Not reported	Not reported	Learning outcomes	Implementation
5	Zhou, Wang; Kim, Yea Jin	GAN	Not reported	Not reported	Feedback	Implementation
6	Zhang et al.	GAN	Not reported	Not reported	Feedback	Implementation
7	Ford et al.	Markov	Not reported	Not reported	Not specified	Not specified
8	Holster	Not specified	Not reported	Not reported	Feedback	Implementation
9	Uzumcu and Acilmis	Not specified	Not reported	Not reported	Feedback	Implementation
10	Subbulakshmi et al.	Not specified	Not reported	Not reported	Creativity, Expression	Emotion, Technical
11	Xu	Not specified	Not reported	Not reported	Feedback	Not specified
12	Fernando et al.	Not specified	Not reported	Not reported	Engagement	Cultural, Emotion
13	Shi	Not specified	Not reported	Not reported	Creativity	Not specified
14	Ferreira et al.	Not specified	Not reported	Not reported	Feedback, Assessment	Emotion
15	Lian	Not specified	Not reported	Not reported	Feedback, Assessment	Emotion
16	Cui	Not specified	Not reported	Not reported	Creativity	Not specified
17	Chu et al.	Not specified	Not reported	Not reported	Creativity	Not specified
18	Novelli and Proksch	Not specified	Not reported	Not reported	Not specified	Emotion
19	Bougueng et al.	Not specified	Not reported	Not reported	Not specified	Not specified
20	Wang and Jiang	Not specified	Not reported	Not reported	Creativity	Not specified
21	Kumar et al.	Not specified	Not reported	Not reported	Creativity	Not specified
22	Dadman et al.	Reinforcement Learning	Not reported	Not reported	Creativity	Not specified
23	Kim et al.	Not specified	Not reported	Not reported	Feedback, Assessment	Bias
24	Yang and Liu	Not specified	Not reported	Not reported	Feedback	Not specified
25	Tubadji et al.	Not specified	Not reported	Not reported	Not specified	Cultural, Emotion
26	Suh et al.	Not specified	Not reported	Not reported	Creativity	Not specified
27	Wang et al.	Not specified	Not reported	Not reported	Feedback	Not specified
28	Tigre Moura and Maw	Not specified	Not reported	Not reported	Assessment	Implementation
29	Lee and Liu	Not specified	Not reported	Not reported	Assessment	Implementation
30	Tanberk and Tükel	CNN	Not reported	Not reported	Creativity	Not specified
31	Dalmazzo et al.	Not specified	Not reported	Not reported	Feedback	Not specified
32	Jin et al.	Not specified	Not reported	Not reported	Feedback	Not specified
33	Chang and Chen	Not specified	Not reported	Not reported	Assessment, Expression	Emotion
34	Gonzalez and McMullen	Not specified	Not reported	Not reported	Assessment, Expression	Emotion
35	Eremenko et al.	Not specified	Not reported	Not reported	Feedback, Assessment	Not specified
36	Samuel and Pilát	LSTM	Not reported	Not reported	Not specified	Not specified

Table 5 (continued)

No.	Authors	AI Model	Accuracy	Latency	Pedagogical Impact	Limitations
37	Giraldo et al.	Not specified	Not reported	Not reported	Learning outcomes	Not specified
38	Carey et al.	Not specified	Not reported	Not reported	Assessment	Not specified
39	Giraldo et al.	Not specified	Not reported	Not reported	Assessment	Not specified
40	Pati et al.	Not specified	Not reported	Not reported	Expression	Not specified
41	Vass	Markov	Not reported	Not reported	Creativity	Not specified
42	Roig et al.	Markov	Not reported	Not reported	Creativity	Not specified
43	Ting et al.	Not specified	Not reported	Not reported	Not specified	Not specified
44	Lopes et al.	Not specified	Not reported	Not reported	Creativity	Not specified
45	Venugopal and Madhusudan	Not specified	Not reported	Not reported	Creativity	Not specified
46	Sabathe, Romain et al.	Autoencoder	Not reported	Not reported	Assessment	Not specified
47	Liu and Ting	Not specified	Not reported	Not reported	Creativity, Feedback	Not specified
48	Cheng et al.	Not specified	Not reported	Not reported	Feedback	Not specified
49	Baudouin	GAN	Not reported	Not reported	Not specified	Not specified
50	Tzimeas and Mangina	Not specified	Not reported	Not reported	Creativity	Not specified

Table 5 includes technical terms related to AI model types. “GAN” refers to Generative Adversarial Networks, which are used to generate new data (e.g., music compositions) by learning patterns from existing datasets. “Markov” models are probabilistic models that predict future states (e.g., musical notes or harmony) based on previous ones. “Autoencoder” is a neural network used for compressing and reconstructing data, often applied in music feature extraction. “LSTM” (Long Short-Term Memory) and “CNN” (Convolutional Neural Network) are deep learning architectures commonly used for sequence modeling and pattern recognition in music performance and composition. In addition, where evaluation metrics, architectural configurations, or training parameters are marked as “Not reported,” this reflects the reporting practices of the original studies rather than omissions in the present review.

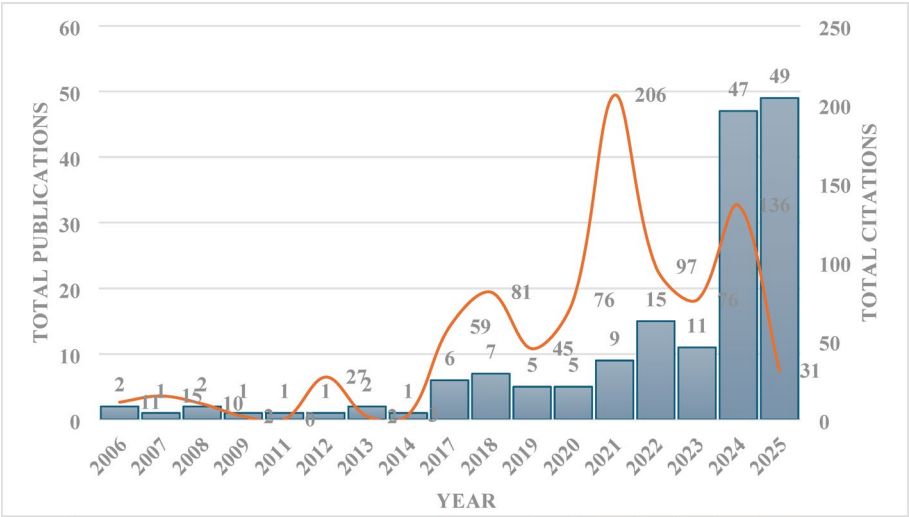


Fig. 3 Total Publications and Citations by Year.
Source: Generated by the author(s) using biblioMagika® [3]

Together, these tables provide a dual-layered analysis, bridging pedagogical insights with computational realities and supporting a more rigorous understanding of how AI is shaping music education. Where quantitative performance indicators (e.g., accuracy, precision, recall, F1-score, or latency) were reported in the source studies, these were

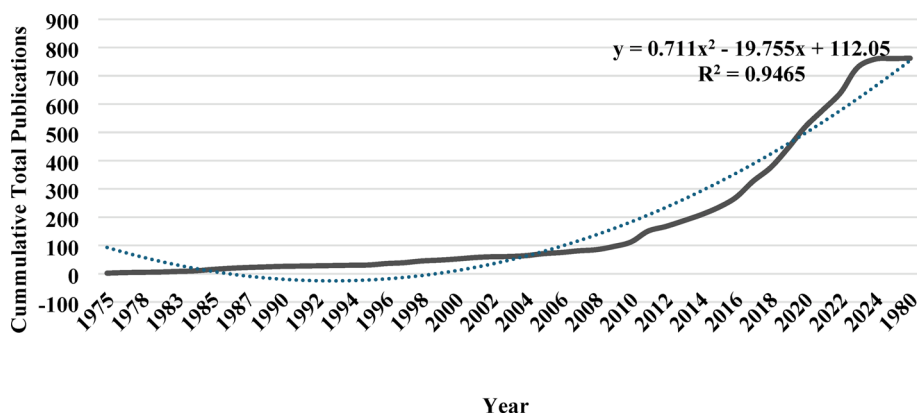


Fig. 4 Publication Growth.
Source: Generated by the author(s) using biblioMagika® [3]

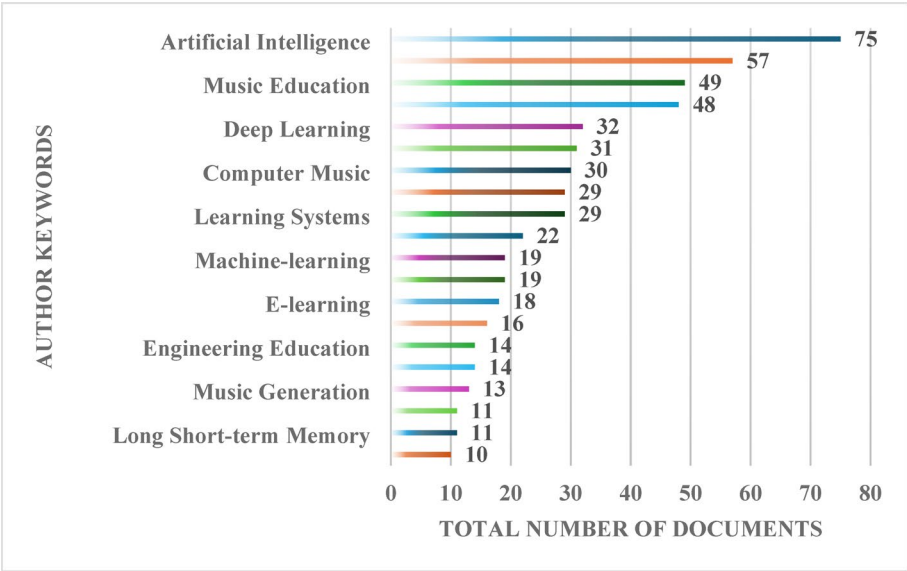


Fig. 5 Top 20 Author keywords by document frequency. The distribution reveals three dominant thematic areas: (1) Core AI Technologies (e.g., 'Artificial Intelligence', 'Deep Learning', 'Machine Learning', 'Neural Networks'), representing the computational foundations; (2) Music Education Domains (e.g., 'Music Education', 'Music Composition', 'Computer Music'), highlighting the application contexts; and (3) Educational Framework (e.g., 'Students', 'Teaching', 'Learning Systems', 'E-learning'), focusing on pedagogical aspects.
Source: Generated by the author(s) using biblioMagika® [3]

tabulated in Table 5. In cases where such metrics were not specified, the cells are labeled as ‘Not reported’ to maintain transparency and consistency across studies.

4 Discussion

This section critically synthesizes the thematic and technical findings presented in Tables 4 and 5, offering a dual-layered perspective on how artificial intelligence (AI) is shaping instrumental music education. By examining algorithmic trends, pedagogical impacts, and implementation challenges, the discussion highlights both the transformative potential and the limitations of AI integration in music learning environments. Thematic and technical insights were derived from a curated subset of 50 highly cited documents, selected from 165 eligible records. This selection ensured analytical depth

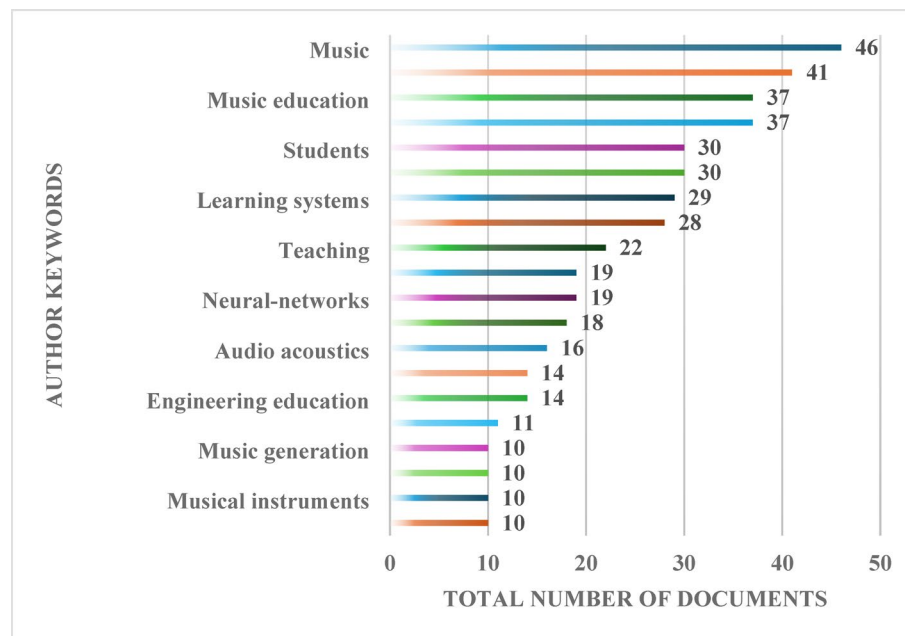


Fig. 6 Top 20 Index keywords by document frequency. The analysis identifies three primary research streams: (1) Technical AI Methods (e.g., 'Deep Learning', 'Neural Networks', 'Long Short-Term Memory'), emphasizing algorithmic approaches; (2) Educational Applications (e.g., 'Music Education', 'Students', 'Teaching', 'Learning Systems'), covering instructional contexts; and (3) Musical Content (e.g., 'Music Composition', 'Computer Music', 'Musical Instruments'), focusing on creative and performance domains.

Source: Generated by the author(s) using biblioMagika® [3]

and relevance across pedagogical and computational domains, as reflected in the Results section.

4.1 Comparative analysis of AI algorithms across educational contexts

A critical analysis of the reviewed literature reveals that the performance and suitability of AI algorithms vary significantly across different educational contexts. For music composition, generative models such as LSTMs and Transformers demonstrate superior capabilities in learning long-term structural dependencies and generating stylistically coherent pieces [38, 56]. More recent large-scale models like MusicLM [2] have further advanced this domain by generating audio from text descriptions, offering powerful tools for creative exploration.

However, a significant pedagogical limitation of these sophisticated models is their tendency to function as 'black boxes,' providing limited explanatory feedback on their compositional choices [105, 121]. This lack of transparency hinders a student's ability to understand the underlying music theory, transforming the AI from a tutor that fosters analytical skills into an oracle that merely provides outputs [144]. The integration of Explainable AI (XAI) is therefore crucial for building trust, ensuring accountability, and unlocking the full educational potential of these tools by making their reasoning accessible to learners [105, 109]. This gap underscores a critical frontier for development in AI-driven music pedagogy.

This inherent limitation of deep generative models underscores the critical need for Explainable AI (XAI) in creative educational domains. XAI is crucial for enhancing transparency, trust, and understanding of AI-driven decisions, which is particularly vital in educational settings where interpretability directly impacts learning outcomes and

user engagement [105, 112]. By providing clear explanations of AI processes, XAI can demystify the 'black box' and help build the trust necessary for effective AI integration in pedagogy [41, 59], addressing a key research gap we identify later in this discussion. In contrast, Genetic Algorithms (GAs) inherently offer a more transparent, exploratory approach to composition by evolving solutions based on explicitly defined fitness functions [107, 142]. This framework models a 'combination-theory of creativity,' making the process interpretable and excellent for teaching compositional principles through experimentation [33, 34]. However, the key limitation of GAs remains their computational inefficiency, rendering them unsuitable for real-time applications where deep learning models excel.

For performance assessment, models excelling in pattern recognition from audio signals are paramount. CNNs are highly effective for tasks like identifying specific techniques (e.g., violin bowing) from spectral features due to their prowess in feature extraction [116]. LSTMs and Bi-LSTMs, with their ability to model temporal sequences, achieve high accuracy in evaluating pitch and rhythm over time [29, 46]. The most robust performance systems often employ hybrid architectures, such as CNN-LSTM networks, which combine spatial feature detection with temporal modeling to provide comprehensive feedback on technique and execution. The primary challenge in this domain is achieving low-latency, real-time feedback, a metric rarely reported in the literature (see Table 5).

In the context of pedagogy and tutoring, Transformer-based models like ChatGPT show immense promise for generating instructional content, answering student queries, and facilitating music theory discussions [52, 145]. Their strength lies in natural language understanding and task adaptation. However, their effectiveness is constrained by the risk of generating plausible but incorrect information ("hallucinations") and a lack of deep, contextualized understanding of individual student progress, an area where Reinforcement Learning (RL) holds potential for creating truly adaptive learning pathways [28, 131].

Recent applications demonstrate the growing sophistication of generative AI in educational contexts. For instance, Arnecke et al. [7] developed an interactive question-answer game using generative AI to help students compose in historical styles, promoting both stylistic knowledge and creativity through human-machine collaboration. Similarly, Wang et al. [133] introduced a user-guided framework combining interactive evolutionary computation with generative adversarial networks for personalized music synthesis, emphasizing the pivotal role of humans in creative music generation. Aligned with emerging higher education trends, recent research identifies adaptive learning systems, AI-driven pedagogical models, and deep learning as central to modern learning environments [77]. These broader trends provide context for the application of such technologies within specialized domains like music education. As a review study, this manuscript does not implement or benchmark deep learning models directly, instead, it synthesizes commonly reported architectural patterns and application-specific uses of AI models across the literature.

4.2 Algorithmic trends and pedagogical domains

Although this review does not introduce new experimental results, existing studies report a range of quantitative performance indicators for AI-based music education

systems. Performance assessment models employing CNN, LSTM and Bi-LSTM-based architectures frequently report classification accuracies exceeding 90%, particularly for tasks involving pitch accuracy, rhythmic alignment, and gesture recognition (e.g., [29, 46]). However, evaluation practices remain highly inconsistent across literature. Base-line comparisons, ablation studies, and system-level metrics such as response latency and real-time feedback stability are rarely reported. As reflected in Table 5, this inconsistency represents a broader methodological gap in the field rather than an omission of the present review, underscoring the need for standardized evaluation protocols that integrate technical metrics with pedagogical relevance.

The integration of artificial intelligence (AI) into instrumental music education presents both exciting opportunities and complex challenges. The thematic synthesis in Table 4 reveals a strong emphasis on composition-focused studies, with deep learning models such as CNN, LSTM, and GAN frequently applied to generative tasks [56, 116]. This aligns with the introduction's observation that AI excels in automating symbolic generation [18]. However, performance-related studies, though fewer, demonstrate nuanced applications in gesture recognition and tonal feedback [29, 43], suggesting a methodological divide between symbolic and embodied learning. Table 5 complements this by showing limited reporting of accuracy and latency, indicating a gap in technical transparency across pedagogical domains.

4.3 Teaching effectiveness: strengths and gaps

AI-powered platforms, such as adaptive practice tools and generative music applications, have shown promising results in improving students' technical accuracy. These tools offer real-time feedback on elements like pitch, rhythm, and articulation, which supports more efficient and independent practice [51, 74, 75, 110]. These findings align with educational theories that emphasize the importance of repetition and immediate feedback in skill acquisition [57, 58]. However, AI remains limited in its ability to support the expressive and creative aspects of music learning. Its lack of cultural awareness and emotional sensitivity can hinder the development of artistic interpretation and originality [4, 78, 101, 138, 139].

4.4 Pedagogical impact and expressive limitations

Creativity and feedback emerge as dominant pedagogical impacts across both tables. Studies such as Moura and Maw [118] and Novelli and Proksch [88] highlight AI's role in fostering novel musical outputs yet caution against emotional detachment and cultural insensitivity. This tension is echoed in Xu et al. [138], who critique AI's inability to capture artistic nuance. While Bi-LSTM and CNN-LSTM models report high accuracy in performance assessment [29, 46], their pedagogical value remains constrained by limited emotional expressiveness and interpretive depth. This supports the argument by Liu and Guo [70] that AI tools must evolve beyond technical precision to support holistic musical development. Hence, building on recent empirical evidence, Naseer et al. [83] convincingly demonstrated that deep learning-based adaptive platforms substantially enhance student outcomes, engagement, and satisfaction in higher education settings.

4.5 Feedback systems and assessment balance

AI offers detailed, data-driven insights into student performance, allowing for precise evaluation of technical skills such as timing and intonation [28, 73]. While this can enhance the objectivity of assessments, there is a risk that over-reliance on quantitative data may overlook important qualitative aspects of musicianship, such as emotional expression and interpretive nuance. A more balanced approach that combines AI-generated metrics with human judgment is essential to ensure a holistic evaluation of musical ability [53, 115]. Table 5 shows that most studies lack latency metrics, which are critical for real-time feedback in music education.

4.6 Ethical and cultural considerations

Ethical concerns such as dataset bias and cultural representation are recurrent across both tables. Tubadji et al. [120] and Fernando et al. [37] emphasize the limitations of Western-centric datasets, while advocate for culturally adaptive tools. Table 5 reinforces these concerns, with emotion and bias frequently cited as limitations. The lack of diversity in training data and the absence of explainability mechanisms highlight the need for inclusive and transparent AI design [6, 97]. Drawing from advancements in healthcare AI, [83, 84, 86] proposed a robust GAN-IoT framework that significantly improves predictive accuracy and privacy, offering a transferable model for ethical AI integration in educational contexts. These findings echo Huang and Zheng [55], who argue that ethical transparency must be embedded in AI system architecture. The ethical dimensions of generative AI in music education are becoming increasingly salient. Recent interview studies with musicians reveal complex tensions between appreciation for AI's potential to increase productivity and independence, and concerns about job displacement, copyright issues, and exacerbation of existing inequities [50]. These findings highlight the need for careful consideration of justice and fairness in generative AI development and deployment, suggesting that different types of AI use, from assistant to replacement, carry distinct ethical implications that must be addressed in educational settings [22, 23].

4.7 Technology access and equity

Access to AI technologies in music education is not evenly distributed. Advanced AI systems often require significant financial and technical resources, which can be a barrier for schools and communities with limited funding [32, 123]. Additionally, many educators are not adequately trained to integrate AI tools into their teaching, which can result in underutilization or ineffective implementation [5, 114]. Hybrid models, where AI supports rather than replaces human instruction, offer a promising solution [145]. These models preserve relational learning while enhancing accessibility [27, 87].

4.8 Identification of concrete research gaps

This review identifies several concrete and pressing research gaps that warrant immediate attention. Furthermore, research on user acceptance indicates that perceived risk significantly influences pre-service music teachers' intention to use generative AI, highlighting the need for addressing ethical concerns in implementation strategies [49].

1. Explainable AI (XAI) for Creative Feedback: There is a critical need to move beyond "black box" AI systems in creative domains. Future research should develop XAI

frameworks that can articulate the rationale behind compositional choices or performance critiques, thereby enhancing the pedagogical value of AI feedback and fostering deeper conceptual understanding [97].

2. **Culturally Adaptive and Linguistically Diverse Datasets:** The pervasive bias in Western-centric training data limits the global applicability of AI music tools. A significant gap exists in creating large-scale, multimodal datasets that encompass diverse musical traditions, genres, and languages, which is a prerequisite for developing culturally sensitive AI systems [37, 120]. This limitation also reflects the geographic imbalance identified in the Results section, where Southeast Asia and Africa remain significantly under-represented in AI and music education research.
3. **Long-Term and Meta-Analytic Studies on Pedagogical Impact:** While short-term efficacy studies are abundant, there is a pronounced lack of longitudinal research examining the sustained impact of AI tools on student motivation, creative self-efficacy, and the development of a personal artistic voice [25, 71]. Future research should employ longitudinal designs and meta-analytic synthesis to evaluate the long-term pedagogical effects of AI integration across diverse educational settings. Such studies would provide evidence-based insights into how AI influences learning trajectories, creativity development, and student engagement over extended periods.
4. **Seamless Multimodal Integration:** Current systems often operate on a single data modality (e.g., audio or symbolic). A key frontier is the development of integrated AI architectures that can simultaneously process and correlate audio, visual (posture, gesture), and symbolic music data to provide a holistic assessment of a student's performance [134]. Emerging frameworks like MusicAI, which combines cross-modal transformers with variational autoencoders to generate emotionally resonant music based on textual prompts, represent promising steps toward truly integrated multimodal systems [126].
5. **Affective Computing and Emotional Responsiveness in AI Music Education:** While AI models have shown strong technical proficiency, their ability to perceive, interpret, and respond to emotional cues remains limited. Future research should explore affective computing approaches that integrate emotion recognition, expression modeling, and user feedback loops. Such systems could enable AI tutors and performance evaluators to adapt not only to technical precision but also to emotional expressiveness and artistic intent, thereby supporting more human-centered and empathetic learning experiences [76, 91, 93, 138, 139, 141].

4.9 Future research needs

Table 5 suggests a lack of multimodal integration and explainability features. Future research should explore culturally responsive datasets, multimodal feedback systems, and ethical frameworks that align with pedagogical goals [44, 96]. Studies exploring mood-based composition using LSTM networks demonstrate AI's potential to bridge computational precision with emotional depth, though further work is needed to fully realize this potential in educational contexts [94]. To support future readiness, Naseer et al. [84] proposed an industry-enriched project-based learning framework that significantly improved student employability and engagement, which could inform music education curriculum design. Longitudinal and meta-analytic studies could further validate the enduring pedagogical impact of AI-based interventions in music education.

AI in Music Education: A Meta-Framework Linking AI Methods, Pedagogical Models, and Educational Applications

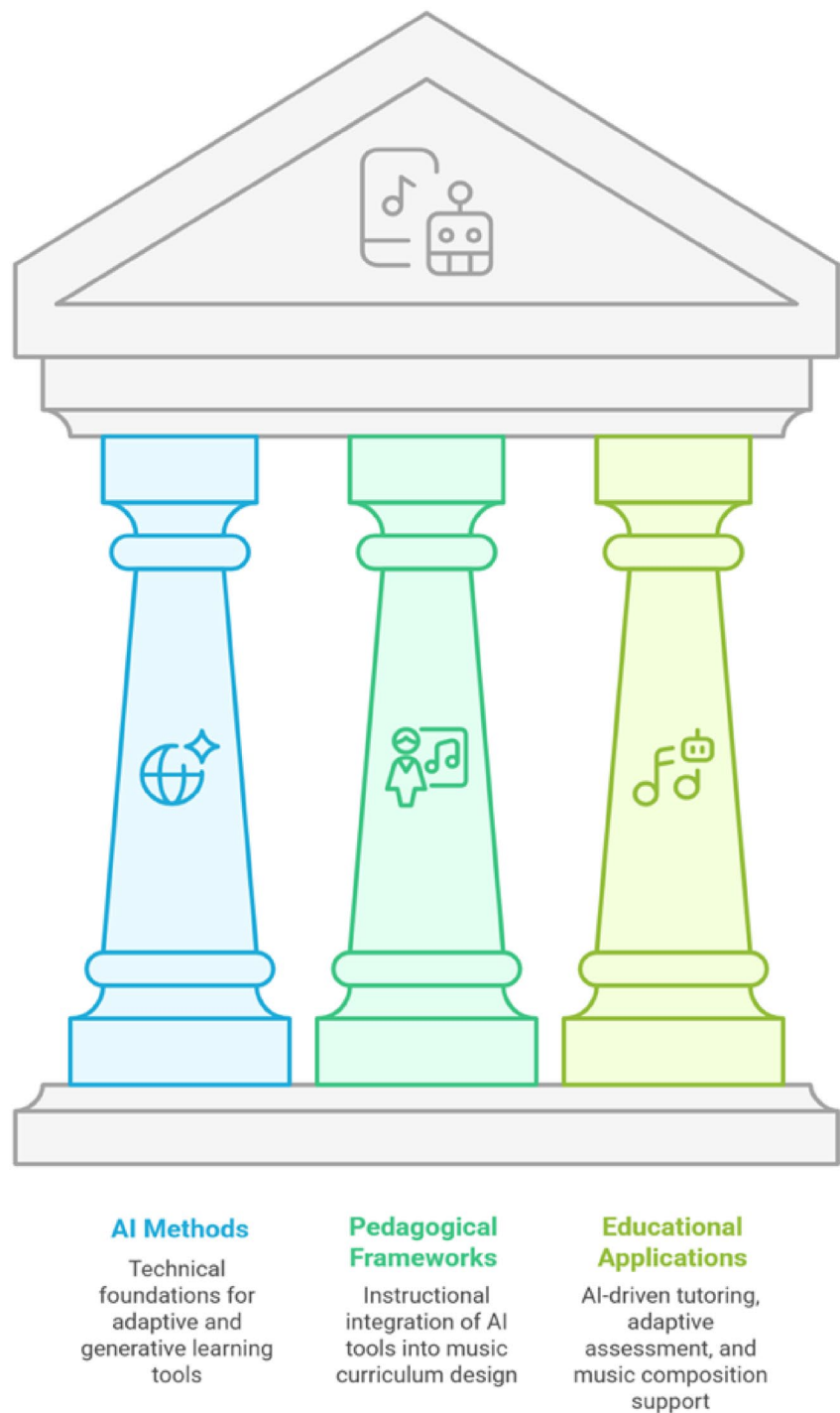


Fig. 7 (See legend on next page.)

(See figure on previous page.)

Fig. 7 Conceptual meta-framework illustrates the foundational relationship between AI methods, pedagogical models, and educational applications in music education. Each pillar represents a core dimension of AI integration: (1) AI Methods: technical foundations for adaptive and generative learning tools; (2) Pedagogical Frameworks: instructional integration of AI tools within models such as TPACK, SAMR, and Bloom's taxonomy; and (3) Educational Applications: AI-driven tutoring, adaptive assessment, and composition support. Together, these dimensions provide a holistic understanding of AI's pedagogical role in music education. Source: Generated by authors using Napkin AI.

4.10 Integrating pedagogical frameworks

While the introduction references pedagogical models, the literature lacks integration of frameworks such as SAMR, TPACK, and Bloom's taxonomy. Bauer [13] and Rabu et al. [95] demonstrate how these models can guide AI adoption in music education. Incorporating these frameworks can enhance instructional design and support higher-order learning outcomes. For example, SAMR can help educators evaluate the transformative potential of AI tools, while TPACK offers a lens to assess the interplay between content, pedagogy, and technology. Extending pedagogical innovation, Naseer et al. [85] introduced a gamified AI framework that markedly boosted learner motivation and mastery-level performance, aligning seamlessly with SAMR and Bloom's taxonomy in adaptive learning environments.

To visualize these relationships, Fig. 6 below presents a conceptual meta-framework representing AI in music education as a three-pillar model. Each pillar denotes a foundational dimension: AI methods, pedagogical frameworks, and educational applications, under a unified educational structure. This framework illustrates how technical innovations (e.g., DL, RL, GAN models) interface with pedagogical theories (TPACK, SAMR, Bloom's taxonomy) to support practical outcomes such as tutoring, adaptive assessment, and composition. Visualization underscores the balance between technological sophistication and pedagogical intentionality in fostering effective AI integration (Fig. 7).

4.11 Conclusion of discussion

Tables 4 and 5 collectively offer a comprehensive view of AI's role in music education. Thematic insights reveal pedagogical strengths, while technical summaries expose gaps in transparency and equity. Bridging these dimensions requires interdisciplinary collaboration, ethical design, and pedagogical sensitivity to ensure that AI enhances rather than diminishes the humanistic essence of music learning.

5 Conclusion

The integration of artificial intelligence into instrumental music education represents a paradigm shift in how musical skills are taught, assessed, and experienced. This review has critically examined the intersection of AI technologies and pedagogical frameworks, revealing both the transformative potential and the nuanced limitations of current implementations. While AI systems have demonstrated efficacy in enhancing practice efficiency, personalization, and objective assessment, their broader educational value depends on thoughtful integration within culturally responsive, ethically sound, and pedagogically grounded frameworks. The dual-layered analysis presented in this review, spanning algorithmic capabilities and instructional impact underscores the need for interdisciplinary collaboration between educators, technologists, and policymakers. Frameworks such as TPACK, SAMR, and Bloom's taxonomy offer valuable lenses for aligning AI tools with cognitive and instructional goals, ensuring that technological

innovation complements rather than displaces human creativity and relational learning. Looking ahead, future research should prioritize the development of hybrid instructional models that combine AI's analytical precision with the expressive, interpretive, and cultural dimensions of music education. By adopting a human-centered and critically reflective approach, the field can harness AI not merely as a tool for automation, but as a catalyst for deeper, more inclusive, and more meaningful musical learning experiences. One limitation of this review is the exclusive use of the Scopus database, as the authors did not have institutional access to Web of Science (WoS) for cross-database validation. However, Scopus has been shown in previous comparative studies to offer broad and reliable coverage, supporting the robustness of the dataset.

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Author contributions

C.A.N.M. conceptualized the study, led the literature review, and coordinated the overall manuscript development. A.R.M.N. contributed to the design of the review framework and analysis of educational trends. M.R.S. supported data extraction and synthesis, and contributed to the interpretation of findings. H.F.H. provided technical expertise in data management and contributed to the methodological section. S.A.S. contributed to the comparative analysis of international perspectives and helped draft the discussion. R.H. assisted in the review of literature and data coding. S.J. provided critical revisions. All authors reviewed and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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